Do Tabular Foundation Models Pay Off for Churn? A Profit-, Calibration- and Tuning-Aware Re-evaluation

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Abstract. Tabular foundation models (TFMs) predict from a labelled context in a single forward pass. A recent benchmark reported that they decisively outperform tree ensembles for customer churn, but it compared them with untuned gradient-boosted trees (GBDTs) and scored only ranking metrics, while a retention campaign is judged in money. We re-evaluate TabPFN-2.5, TabPFN-3.5, TabPFN-3.5-Fast and TabICL v2 against Optuna-tuned LightGBM, XGBoost and CatBoost on five public telecom churn datasets. Every model is scored on expected maximum profit (EMP), e-Profits, calibration and scoring time, over five seeds with paired-bootstrap intervals. Only TabPFN-3.5 earns a significant EMP gain on more than one dataset (Orange 17.8%, Cell2Cell 1.9%, Iranian 7.5%), and it is never significantly worse. TabICL v2, the earlier benchmark's leader, is significant only on Iranian and underpredicts Orange churn by 2.3 points on a 7.3% base rate. Profit metrics reorder models mainly when they are statistically tied. A class-balanced context lowers TabPFN-3.5's Orange gain rather than raising it. On the earlier benchmark's headline dataset, tuning the baseline cuts TabICL v2's PR-AUC lead from 6.1 to 4.0 points without closing it. Scoring cost is minor; the licence of TabPFN-3.5's weights, which forbids production use, is the binding constraint.

Keywords: Customer churn · Tabular foundation models · Expected maximum profit · Calibration · Benchmarking

1 Introduction

Churn models exist to choose whom a retention campaign contacts. A model is useful to an operator to the extent that the customers it ranks highest are the ones worth an incentive, so the natural yardstick is campaign profit rather than classification accuracy. Verbraken et al. made this precise with the expected maximum profit (EMP) criterion [32], and later work showed that selecting churn models by AUC instead of profit leaves money on the table [31,13,17].

Tabular foundation models change what a churn model is. TabPFN [12,20,22] and TabICL [24,25] are pretrained on synthetic tables and predict a new customer by conditioning on the labelled training set as context, with no fitting

and no hyperparameter search. On general benchmarks they now lead or match tuned GBDTs [10,25,22]. For churn specifically, Seyedzadeh and Karimi [26] benchmarked six TFMs on nine Kaggle datasets. They concluded that TFMs "collectively and decisively outperform" classical, deep and tree-ensemble models, with TabICL v2 ahead of XGBoost by up to 9.23 PR-AUC points.

That study was the first to bring TFMs to churn, and it leaves three questions open. Its XGBoost and random forest used library-default hyperparameters [26, Sect. 3.2], so part of the reported gap may be a tuning gap. Its metrics were F1, ROC-AUC, PR-AUC and log loss; its financial estimate scaled the PR-AUC gap by a retention rate rather than evaluating a campaign. Finally, its own discussion reads TabICL v2's lead as "operational ... rather than as a statistically decisive metric lead" [26, Sect. 4.2], which leaves the size of the lead unsettled.

We re-evaluate the question with the baselines tuned and the decision made explicit. Four TFMs (TabPFN-2.5, TabPFN-3.5, TabPFN-3.5-Fast, TabICL v2) face Optuna-tuned LightGBM, XGBoost and CatBoost, an explainable boosting machine and logistic regression on five public telecom datasets, three of them from real operators. Every model is scored on ranking metrics, EMP, e-Profits, a fixed-threshold campaign policy, calibration and scoring time, over five seeded splits with paired-bootstrap intervals. Our findings:

1. **Only TabPFN-3.5 pays off consistently.** Against the best tuned GBDT its EMP gain is significant on 3 of 5 datasets and never negative beyond noise; TabICL v2 and TabPFN-2.5 are significant only on Iranian. On the two fictional datasets, Telco and Maven, no TFM differs significantly from the GBDTs (Sect. 4.1).
2. **Profit metrics reorder ties, not clear winners.** Where one model leads clearly (Orange, Cell2Cell), the EMP winner equals the AUC winner in every seed; where models are within noise, the EMP winner differs in up to 4 of 5 seeds (Sect. 4.2).
3. **Calibration matters for one model on one dataset.** Raw TabICL v2 underpredicts Orange churn by 2.3 points and loses 15.7% of attainable campaign profit at a fixed threshold; elsewhere recalibration changes that loss by at most 1.4 points (Sect. 4.3).
4. **A balanced context does not help when the full training set fits.** It lowers TabPFN-3.5's Orange gain from 17.8% to 4.5% (Sect. 4.4).
5. **Tuning shrinks but does not erase the headline gap.** On the earlier benchmark's E-Commerce dataset, TabICL v2's PR-AUC lead falls from 6.1 points over default XGBoost to 4.0 over tuned CatBoost, and remains significant (Sect. 4.5).

We also report scoring throughput and licensing (Sect. 4.6), which together decide whether a gain can be deployed at all. The code, every per-split prediction and the table generator are released (Sect. 7).

Table 1. Closest prior studies. Tuned: baselines hyperparameter-tuned. Calibration: analysed beyond reporting log loss. Tests: uncertainty or significance reported for differences between models.

Study	Domain	TFMs	Tuned	Profit metric	Calibration	Tests
Manzoor et al. [17]	churn	no	yes	e-Profits, EMP	no	yes
Baesens et al. [8]	credit risk	yes	yes	no	yes	yes
Seyedzadeh & Karimi [26]	churn	yes	no	no	log loss only	no
This work	churn	yes	yes	EMP, e-Profits, policy	yes	yes

2 Related Work

Profit-aware churn evaluation. Verbraken et al. defined EMP as the expected best campaign profit when the probability that a contacted churner accepts the offer is uncertain and Beta distributed [32]. Verbeke et al. used profit to compare churn classifiers in telecom [31], and Höppner et al. optimised EMP directly when growing decision trees [13]. Manzoor et al. proposed e-Profits, which replaces EMP's population-level customer value with a per-customer value derived from Kaplan–Meier retention, and found that it reranks six classical models on IBM Telco and Maven [17]. None of these studies includes a foundation model.

Tabular foundation models. TabPFN v2 [12] established prior-fitted transformers for tables up to roughly 10k rows; TabPFN-2.5 extended this to 50k rows [20], and TabPFN-3 and 3.5 to a million rows with a cached context for fast scoring [21,22]. TabICL separates column and row attention to scale in-context learning [24]; TabICL v2 reports beating a tuned and ensembled RealTabPFN-2.5 on TabArena and TALENT without tuning [25]. On TabArena, GBDTs remain strong on larger datasets while TFMs excel on small ones [10], and TabPrep shows that feature engineering, absent from most benchmarks, changes these comparisons [30]. TabPFN's predictive probabilities are reported to be well calibrated out of the box [28], while ensembling TFMs can damage calibration [27].

TFMs in business prediction. The closest analogue to our study is the credit-risk review of Baesens et al. [8]. Across 14 probability-of-default datasets TabPFN-3 ranks first, but its margin over the strongest GBDT is narrow and "driven by the latest TabPFN and TabICL generations rather than by the model class as a whole". TabPFN-3 also underpredicts the mean default rate. Tanna et al. found that, under heavy imbalance, building a class-balanced context adds 3 to 4 AUC points over uniform sampling for subsampled contexts of 1k to 50k rows [29]; V4FinBench reports a similar benefit of imbalance-aware context construction for bankruptcy prediction [16]. For churn, the only TFM benchmark is [26]. Table 1 sets the closest studies against ours.

Table 2. Datasets after preprocessing. Dup.: exact duplicate feature rows removed. e-Prof.: tenure and monthly revenue available for e-Profits.

Dataset	Source	Rows	Feat.	Churn %	Dup.	e-Prof.	Role
Telco [3]	IBM sample data (fictional)	7,003	19	26.4	40	yes	main
Maven [5]	Maven Analytics (fictional)	6,589	34	28.4	0	yes	main
Cell2Cell [4]	US wireless carrier	51,047	56	28.8	0	yes	main
Iranian [6]	Iranian telecom operator	2,836	15	15.6	314	no	main
Orange [11]	Orange, KDD Cup 2009	50,000	202	7.3	0	no	main
E-Commerce [1]	Kaggle, origin unstated	5,074	18	16.6	556	no	replication

3 Benchmark Design

3.1 Datasets

Table 2 lists the six datasets. Five form the main benchmark. Cell2Cell, Iranian and Orange come from real operators: Cell2Cell from a US wireless carrier [4], Iranian from an Iranian operator through the UCI repository [6], and Orange from the KDD Cup 2009 churn task [11], which we load from OpenML (dataset 1112). IBM Telco [3] and Maven [5] are fictional teaching datasets; we keep them because prior churn studies use them, including both [17] and [26], and because a gap between real and fictional data would itself be informative. E-Commerce [1], the dataset on which [26] reported its largest gain, serves only as a replication check (Sect. 4.5) and is excluded from all cross-dataset summaries.

All models share one preprocessing pipeline. Categorical columns are ordinal-encoded with missing values left as missing, and columns that are more than 99% missing or constant are dropped. For Maven we drop customers whose status is "Joined", whose outcome is not yet observed, and the two columns that record the churn reason after the fact. We remove exact duplicate feature rows, keeping the first: an in-context learner can otherwise look a test customer up in its own context. This removed 40 rows (0.6%) from Telco, 314 (10.0%) from Iranian and 556 (9.9%) from E-Commerce; the other datasets had none.

3.2 Models

Foundation models. TabPFN-2.5, TabPFN-3.5 and TabPFN-3.5-Fast run through the `tabpfn` package (v9.2.0) with default settings and the training-size guardrail disabled; TabICL v2 runs through `tabicl` (v2.2.0, checkpoint of 2026-02-12). Each receives the full training split as its context, at the natural churn rate. None is tuned, which is how these models are meant to be used.

Tuned baselines. LightGBM [15] and XGBoost [9] are tuned with 40 trials of Optuna's TPE sampler [7], and CatBoost [23] with 25, each under a 20-minute cap. Trials are scored by validation AUC on an inner 80/20 split of the training data with early stopping; the best configuration is then refit on the whole training split with the selected number of trees. We also report an explainable boosting machine (EBM) [18] and logistic regression with one-hot encoding and median imputation, both at defaults.

3.3 Protocol

For each dataset and each of five seeds we draw a stratified 60/20/20 split into training, calibration and test sets. All models see identical splits, which the analysis verifies by comparing test indices across runs. Models are fitted (or given context) on the training split. The calibration split is used only to fit post-hoc recalibration, and every reported metric is computed on the test split. Uncertainty for a difference between two models is a paired percentile bootstrap: we resample test customers 500 times per seed, recompute the metric for both models on each resample, and pool the 2,500 differences across seeds. We call a difference significant when its 95% interval excludes zero. Each TFM is compared with the best tuned GBDT for that dataset and metric, chosen by mean over seeds, which favours the baseline. GBDTs ran on Kaggle CPU sessions and TFMs on a single NVIDIA T4 (16 GB).

3.4 Metrics

Ranking. ROC-AUC, PR-AUC and lift in the top decile.

EMP. For a campaign that contacts the fraction of customers with the highest scores, a contacted churner accepts the offer with probability γ and is then worth $\mathrm{CLV} - \delta - \phi$; a churner who declines costs the contact cost ϕ; a contacted non-churner takes the incentive and costs $\delta + \phi$. EMP is the expected profit per customer at the profit-maximising cutoff, averaged over $\gamma \sim \mathrm{Beta}(6, 14)$ [32]. We use the standard parameters $\mathrm{CLV} = 200$, $\delta = 10$ and $\phi = 1$, and integrate over γ on a 200-point grid. Because EMP chooses the best cutoff for each model, it depends only on the ranking and cannot reward calibration.

e-Profits. Following [17], each customer's value is $\mathrm{CLV}_i = R_i M/(1 - \min(r_i, 0.995))$, with monthly revenue R_i, margin $M = 0.3$, and one-month retention $r_i = S(T_i + 1)/S(T_i)$ from a Kaplan–Meier curve S [14] fitted on the training split at the customer's tenure T_i. The offer costs $0.1\,\mathrm{CLV}_i$ and the contact $\max(5, 0.3 \cdot \mathrm{offer})$. A contacted churner yields CLV_i minus both costs; a contacted non-churner yields minus both costs. We report e-Profits when contacting customers with predicted probability at least 0.5, as in [17], and when contacting the top 20%. It is defined only where tenure and revenue exist (Table 2).

Campaign policy and calibration. An operator who acts on probabilities contacts customer i when its expected profit under the EMP cost model is positive at the mean acceptance rate $\bar{\gamma} = 0.3$. Unlike EMP, this policy uses the probability itself, so miscalibration costs money. We report its regret: the profit lost relative to the best cutoff on the same ranking, as a share of that best profit, as the median over a 3×3 grid of $\mathrm{CLV} \in \{100, 200, 500\}$ and $\delta \in \{5, 10, 20\}$. Calibration is measured by expected calibration error (ECE, 15 equal-width bins), Brier score, and calibration-in-the-large: the observed churn rate minus the mean predicted probability, in percentage points, where a positive value means churn is underpredicted. Each model is evaluated raw and after two recalibrations fitted on the calibration split: isotonic regression [33] and Platt scaling [19].

Table 3. Mean PR-AUC and EMP (profit per customer, CLV units) over five seeds; best per column in bold.

	Telco		Maven		Cell2Cell		Iranian		Orange	
Model	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP
Logistic regression	0.650	10.191	0.823	12.694	0.390	8.668	0.786	6.532	0.175	0.422
EBM	0.661	10.168	0.890	13.045	0.463	9.141	0.934	7.614	0.195	0.513
LightGBM	0.657	10.193	0.887	13.059	0.471	9.186	0.943	7.852	0.205	0.564
XGBoost	0.659	10.207	0.887	13.011	0.468	9.183	0.934	7.797	0.201	0.562
CatBoost	0.663	10.137	0.888	13.044	0.469	9.152	0.940	7.762	0.212	0.555
TabICL v2	0.668	10.237	0.890	13.042	0.474	9.224	0.986	8.442	0.213	0.554
TabPFN-2.5	0.667	10.208	0.888	13.078	0.476	9.216	0.985	8.393	0.210	0.558
TabPFN-3.5	**0.668**	**10.266**	**0.899**	**13.112**	**0.499**	**9.360**	**0.990**	**8.470**	**0.241**	**0.666**
TabPFN-3.5-Fast	0.668	10.248	0.895	13.072	0.490	9.299	0.990	8.423	0.230	0.618

Table 4. Each TFM minus the best tuned GBDT: PR-AUC difference in points (×100) and EMP difference relative to the GBDT. Bold with ∗: the 95% paired-bootstrap interval excludes zero. The reference is the best tuned GBDT per dataset and metric, shown as PR-AUC / EMP reference (LGBM: LightGBM, XGB: XGBoost, CatB: CatBoost).

	Telco		Maven		Cell2Cell		Iranian		Orange	
	vs. CatB / XGB		vs. CatB / LGBM		vs. LGBM / LGBM		vs. LGBM / LGBM		vs. CatB / LGBM	
TFM	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP
TabICL v2	+0.5	+0.3%	+0.2	0.0%	+0.4	+0.4%	**+4.3***	**+7.1%***	+0.1	−1.9%
TabPFN-2.5	+0.4	+0.1%	0.0	+0.2%	+0.5	+0.3%	**+4.2***	**+6.6%***	−0.2	−0.8%
TabPFN-3.5	+0.5	+0.5%	+1.0	+0.5%	**+2.8***	**+1.9%***	**+4.7***	**+7.5%***	**+2.9***	**+17.8%***
TabPFN-3.5-Fast	+0.5	+0.4%	+0.7	+0.2%	**+1.9***	+1.2%	**+4.6***	**+7.0%***	+1.8	+9.8%

4 Results

4.1 Do foundation models beat tuned GBDTs?

TabPFN-3.5 is the only TFM with a significant EMP gain on more than one dataset (Table 4). Against the best tuned GBDT it gains 17.8% on Orange, 1.9% on Cell2Cell and 7.5% on Iranian, and it is never significantly worse. TabICL v2 and TabPFN-2.5 are significant only on Iranian, where every boosted and foundation model scores above 0.98 AUC (Appendix, Table 10) and the remaining errors are few. TabPFN-3.5-Fast sits between its parent and TabPFN-2.5.

The Orange result carries the largest effect and the weakest certainty. The 17.8% gain corresponds to an EMP difference whose 95% interval starts at +0.005 per customer, just above zero, because Orange has only 7.3% churners. We therefore read it as evidence of a real gain of uncertain size.

On the two fictional datasets the picture is flat. No TFM differs significantly from the best GBDT on Telco or Maven, on any metric; TabPFN-3.5's EMP gains there are 0.5% and 0.5%. The version also matters more than the family: TabPFN-2.5 and TabPFN-3.5 share an architecture lineage, yet only the latter separates from the GBDTs on Orange and Cell2Cell. This matches the pattern Baesens et al. report for credit risk [8].

Table 5. Agreement between the AUC ranking of all nine models and profit-based rankings: mean Kendall τ over seeds, and seeds in which the top model changes. e-Profits needs tenure and revenue.

	EMP		e-Profits, top 20%		e-Profits, $p \geq 0.5$	
Dataset	τ	top-1 changed	τ	top-1 changed	τ	top-1 changed
Telco	0.61	3/5	0.24	5/5	0.17	5/5
Maven	0.60	4/5	0.37	3/5	−0.10	4/5
Cell2Cell	0.83	0/5	0.46	0/5	0.40	0/5
Iranian	0.80	2/5	–	–	–	–
Orange	0.83	0/5	–	–	–	–

4.2 Do profit metrics change which model wins?

They change it when models are tied and leave it alone otherwise (Table 5). On Orange and Cell2Cell, where TabPFN-3.5 leads clearly, the EMP winner is the AUC winner in all five seeds, and Kendall's τ between the two rankings is 0.83 and 0.83. Telco and Maven behave differently: their top models are within noise of each other, and the EMP winner differs from the AUC winner in 3 and 4 of 5 seeds.

e-Profits at a 0.5 threshold disagrees with AUC much more ($\tau = -0.10$ on Maven). This is expected from its construction rather than evidence of a better ranking: at a fixed threshold the metric also rewards a model whose probabilities happen to sit on the right side of 0.5, so it mixes ranking with calibration. The reranking reported for e-Profits on Telco and Maven [17] is therefore reproduced here, and our bootstrap intervals locate it among models whose ranking metrics are statistically indistinguishable.

4.3 Does calibration cost money?

Mostly it does not. Outside Orange, the fixed-threshold policy loses at most 2.9% of attainable profit for any model without recalibration, and isotonic or Platt scaling changes that loss by at most 1.4 points (Table 6).

Orange is the exception, and one model drives it. Raw TabICL v2 predicts 2.3 percentage points too little churn on a base rate of 7.3%, and the policy built on those probabilities loses 15.7% of attainable profit. Isotonic regression cuts the loss to 8.2% and Platt scaling to 6.0%. TabICL v2 also underpredicts on Cell2Cell (by 1.6 points), as does TabPFN-2.5 (1.8). TabPFN-3.5's calibration-in-the-large stays within ±0.4 points on every dataset, so it is the only TFM here that can be thresholded without recalibration. The tuned GBDTs are close to zero as well, consistent with their fitting to the observed base rate.

4.4 Does a balanced context help?

Under heavy imbalance, a balanced context has been reported to add 3 to 4 AUC points for TFMs [29]. If that held here, our full-context runs would understate

Table 6. Calibration. CITL: observed minus mean predicted churn rate (points; positive = underprediction). Regret: share of best-cutoff profit lost by the fixed-threshold policy (median over the cost grid), raw / isotonic / Platt. The first row per dataset is the best tuned GBDT.

			ECE		Policy regret (%)		
Dataset	Model	CITL	raw	iso.	raw	iso.	Platt
Telco	XGBoost	+0.1	0.030	0.030	1.5	1.4	1.4
	TabICL v2	+0.8	0.029	0.026	1.1	1.6	1.2
	TabPFN-2.5	+0.8	0.027	0.030	1.3	1.7	1.3
	TabPFN-3.5	+0.2	0.029	0.027	1.4	1.7	1.3
Maven	LightGBM	+0.3	0.026	0.025	0.7	0.9	0.9
	TabICL v2	+0.9	0.026	0.026	1.2	1.2	0.9
	TabPFN-2.5	+0.8	0.020	0.030	1.0	1.5	1.2
	TabPFN-3.5	+0.2	0.021	0.022	0.9	1.1	1.1
Cell2Cell	LightGBM	0.0	0.011	0.012	0.5	0.6	0.4
	TabICL v2	+1.6	0.017	0.012	0.5	0.6	0.5
	TabPFN-2.5	+1.8	0.020	0.011	0.6	0.5	0.3
	TabPFN-3.5	−0.4	0.013	0.015	0.5	0.5	0.5
Iranian	LightGBM	+0.6	0.030	0.022	1.8	2.4	1.8
	TabICL v2	−0.3	0.014	0.009	0.9	1.2	1.1
	TabPFN-2.5	−0.3	0.015	0.010	0.8	0.7	0.8
	TabPFN-3.5	−0.3	0.012	0.012	1.0	0.7	0.9
Orange	LightGBM	+0.1	0.009	0.006	7.9	6.0	5.4
	TabICL v2	+2.3	0.024	0.004	15.7	8.2	6.0
	TabPFN-2.5	+0.2	0.008	0.006	8.1	5.9	7.5
	TabPFN-3.5	0.0	0.005	0.005	5.1	3.6	6.1

the TFMs on Orange. We kept every churner in the training split and sampled one or two non-churners per churner, for TabPFN-3.5 and TabICL v2 on Orange and Cell2Cell, on the same splits and seeds.

Balancing hurt (Table 7). TabPFN-3.5's EMP gain on Orange fell from 17.8% with the full context to 8.4% at 1:2 and 4.5% at 1:1, and neither of the latter is significant. TabICL v2 gained nothing on either dataset. Balancing also inflated raw predicted churn: on Orange the 1:1 context overpredicts the churn rate by 35.3 points, an error that isotonic recalibration on the calibration split removes. The difference from [29] is the regime: there, contexts were subsampled to at most 50k rows from larger data, so balancing chose which customers to keep. Here the whole training split fits, and undersampling only discards non-churners.

4.5 Does the earlier headline survive tuning?

We reran our protocol on E-Commerce, where [26] reported TabICL v2 9.23 PR-AUC points ahead of XGBoost, adding a library-default XGBoost to reproduce their baseline (Table 8). The direction survives. TabICL v2 leads default XGBoost by 6.1 points and the best tuned GBDT, CatBoost, by 4.0 points, both significant, and its EMP gain is 4.8%. Tuning raised the GBDT's PR-AUC from 0.906 to 0.927 and so removed about a third of the gap. TabPFN-3.5 leads by more (5.7 points over tuned CatBoost; EMP gain 8.2%). Our protocol differs from theirs (a 60/20/20 holdout over five seeds against five-fold cross-

Table 7. Context construction. Full: the whole training split at its natural churn rate. 1:2 and 1:1: all churners plus two or one sampled non-churners per churner. ΔEMP is relative to the best tuned GBDT (bold $*$: significant). CITL in points, raw.

Dataset	Model	Context	AUC	PR-AUC	EMP	ΔEMP	CITL
Orange	TabPFN-3.5	full (natural rate)	0.747	0.241	0.666	+**17.8**%*	0.0
Orange	TabPFN-3.5	1:2	0.739	0.219	0.611	+8.4%	−21.6
Orange	TabPFN-3.5	1:1	0.736	0.208	0.589	+4.5%	−35.3
Orange	TabICL v2	full (natural rate)	0.725	0.213	0.554	−1.9%	+2.3
Orange	TabICL v2	1:2	0.730	0.209	0.557	−1.1%	−19.9
Orange	TabICL v2	1:1	0.727	0.196	0.525	−7.0%	−35.1
Cell2Cell	TabPFN-3.5	full (natural rate)	0.705	0.499	9.360	+**1.9**%*	−0.4
Cell2Cell	TabPFN-3.5	1:2	0.704	0.498	9.357	+**1.9**%*	−4.4
Cell2Cell	TabPFN-3.5	1:1	0.700	0.491	9.324	+**1.5**%*	−19.1
Cell2Cell	TabICL v2	full (natural rate)	0.689	0.474	9.224	+0.4%	+1.6
Cell2Cell	TabICL v2	1:2	0.689	0.474	9.226	+0.4%	−2.7
Cell2Cell	TabICL v2	1:1	0.686	0.468	9.197	+0.1%	−19.1

Table 8. Replication on E-Commerce. PR-AUC mean (sd) over five seeds; Δ in PR-AUC points versus default XGBoost and versus the best tuned GBDT, and EMP relative to the best tuned GBDT (bold $*$: significant).

Model	PR-AUC	EMP	ΔPR vs default	ΔPR vs tuned	ΔEMP
XGBoost (library default)	0.906 (0.019)	8.090	–	–	–
XGBoost	0.923 (0.019)	8.119	–	–	–
LightGBM	0.922 (0.016)	8.116	–	–	–
CatBoost	0.927 (0.017)	8.119	–	–	–
TabICL v2	0.967 (0.012)	8.505	+**6.1***	+**4.0***	+**4.8**%*
TabPFN-2.5	0.966 (0.012)	8.512	+**5.9***	+**3.8***	+**4.9**%*
TabPFN-3.5	0.984 (0.008)	8.790	+**7.9***	+**5.7***	+**8.2**%*

validation), so we compare default and tuned baselines within our own runs rather than with their numbers. On this dataset, then, TFMs win clearly. The origin of E-Commerce is not documented, so it cannot tell us whether the effect is specific to real or synthetic data.

4.6 Scoring cost and licensing

An in-context learner re-reads its context for every scoring call unless the context is cached, so its cost depends on how scoring is run. We timed scoring of 1k, 10k and 50k customers resampled from the test split, after a warm-up call, on one T4 (Table 9). With the context cached, TabPFN-3.5 scores 10k Orange customers in 45 s against a 30,000-row context, and TabICL v2 with its key-value cache in 17 s, against 0.8 s for a 1,000-tree LightGBM on CPU. A monthly scoring run of a million customers therefore costs about 1.25 T4-hours with cached TabPFN-3.5. That is slow next to a GBDT but small next to the per-customer profit differences in Table 4. Without caching, small batches are very expensive (1,686 s per 10k for TabPFN-3.5 at 1k rows on Orange), and TabPFN-2.5 was impractical at 30k-row contexts.

Table 9. Seconds to score 10k customers on one T4 (LightGBM on CPU), best of 10k and 50k batches. †: only the 1k batch was run because larger ones exceeded our time budget; per-customer time falls with batch size, so this is an upper bound. Context: training split (4,201 / 30,628 / 30,000 rows).

Model	Telco	Cell2Cell	Orange
LightGBM (1,000 trees, CPU)	0.8	0.5	0.8
TabICL v2	2.5	18.8	89.2
TabICL v2, KV cache	–	–	16.7
TabPFN-3.5	8.8	49.4	1,686†
TabPFN-3.5, cached context	8.1	30.5	44.9
TabPFN-3.5-Fast	2.7	12.4	25.9
TabPFN-2.5	9.6	3,065†	10,122†

Licensing constrains deployment more than compute does. The TabPFN-3.5 weights are released for research and evaluation, but "the models, their derivatives, and their outputs cannot be used for commercial or production purposes", including "using model outputs as inputs to internal commercial decision-making" [22, Sect. 4][2]; production use requires a commercial licence. TabICL v2's weights are BSD-3 licensed. An operator choosing between them therefore faces a trade-off our benchmark makes concrete: the model that pays off consistently needs a commercial agreement, and the openly licensed one is not reliably better than a tuned GBDT on these datasets.

5 Discussion

The results support a narrower statement than "foundation models outperform trees for churn". Gains are real on some datasets and absent on others, and three factors explain most of the variation we see. The model version matters more than the model class: TabPFN-3.5 separates from tuned GBDTs where TabPFN-2.5 does not. Baseline tuning removes part of any measured lead, about a third of it on E-Commerce. And the datasets most used in churn research, Telco and Maven, are the ones on which no model can be told apart, so studies built on them can neither confirm nor refute a TFM advantage.

For a practitioner this suggests an order of operations. Tune the GBDT first; it is the baseline that matters, and default settings overstate what a TFM adds. If TabPFN-3.5 still leads on held-out profit and the licence permits it, use it with a cached context. If only openly licensed models are possible, test TabICL v2 on one's own data and recalibrate it before thresholding. A balanced context is not a free improvement when the training set fits.

The finding most likely to change with more evidence is the size of the Orange gain. Its interval is wide, and a stronger GBDT pipeline with feature engineering [30] or a larger tuning budget [10] could narrow it. The finding least likely to change is the calibration pattern, which is consistent across seeds and matches the underprediction reported for credit risk [8].

6 Limitations

Our GBDT tuning is lighter than benchmark practice: 25 to 40 TPE trials, against 200 configurations with post-hoc ensembling in TabArena [10]. Because heavier tuning can only help the baselines, our TFM gains are, if anything, overstated. We did no feature engineering, which recent work shows can shift model comparisons [30].

The evidence rests on three real-operator datasets, and the largest effect, on Orange, has an interval close to zero. We cannot claim that TabPFN-3.5's gain generalises to churn in other industries or to operators with millions of customers, where the context would no longer hold the full training set. The cost model behind EMP and the policy uses the standard population-level parameters [32]; e-Profits gives a customer-level alternative on three datasets only.

We used one splitting protocol (60/20/20 holdout, five seeds) rather than repeated cross-validation, one GPU type, and the default inference settings of each TFM. We did not evaluate the API-only TabPFN-3.5-Plus and -Thinking variants, or the other TFMs covered by [26].

7 Conclusion

Measured with tuned baselines and in money, tabular foundation models pay off for churn selectively. TabPFN-3.5 improves campaign profit significantly on three of five telecom datasets and never loses beyond noise, but its weights cannot be used in production without a commercial licence. TabICL v2 and TabPFN-2.5 are rarely distinguishable from a tuned GBDT, and TabICL v2 needs recalibration before its probabilities can drive a campaign. Profit metrics confirm clear winners and reshuffle ties. We release the pipeline and every per-split prediction so that these conclusions can be re-checked as models and licences change.

Data and Code Availability

The version of the code and results described here is archived on Zenodo as release v1.1 (`https://doi.org/10.5281/zenodo.23696051`) and maintained at `https://github.com/ihpolash/tfm-churn-profit-audit`. The repository contains the prediction and analysis code as scripts and as notebooks, the Kaggle kernel definitions that ran them, the per-split predictions and split indices of all 315 runs, the analysis outputs, and the generator and audit for every table and number in this paper. The datasets are public (Table 2) but are not redistributed, because several are licensed for non-commercial or attributed use only. The labels needed to recompute metrics from the predictions are regenerated from the original sources by the released code.

Declaration on the Use of Generative AI

The author used an AI assistant (Claude, Anthropic) for implementation and execution of the experiment code, for literature search and reference checking, for table generation, and for drafting and editing this manuscript. The research question was chosen by the author; the benchmark design, evaluation protocol and interpretation of the results were developed in iterative discussion with the assistant, and every design decision and conclusion was reviewed and approved by the author. All numerical results were produced by the released code. Every table and in-text number is generated by a released script from the released analysis outputs, which are computed from the released per-split predictions, and an audit notebook re-derives them before submission. Every reference was checked against its primary source or an authoritative index. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

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A Additional Results

Table 10. Mean ROC-AUC over five seeds; best per column in bold.

Model	Telco	Maven	Cell2Cell	Iranian	Orange
Logistic regression	0.844	0.917	0.622	0.941	0.703
EBM	0.845	0.942	0.684	0.983	0.725
LightGBM	0.844	0.941	0.687	0.987	0.731
XGBoost	0.845	0.940	0.687	0.985	0.731
CatBoost	0.845	0.941	0.685	0.986	0.726
TabICL v2	0.848	0.942	0.689	0.996	0.725
TabPFN-2.5	0.848	0.942	0.691	0.996	0.731
TabPFN-3.5	**0.848**	**0.947**	**0.705**	**0.997**	**0.747**
TabPFN-3.5-Fast	0.848	0.945	0.700	0.997	0.741